Electrical Circuit (1)

Introduction (week3 class1)

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Chapter 3 Methods of Analysis 81

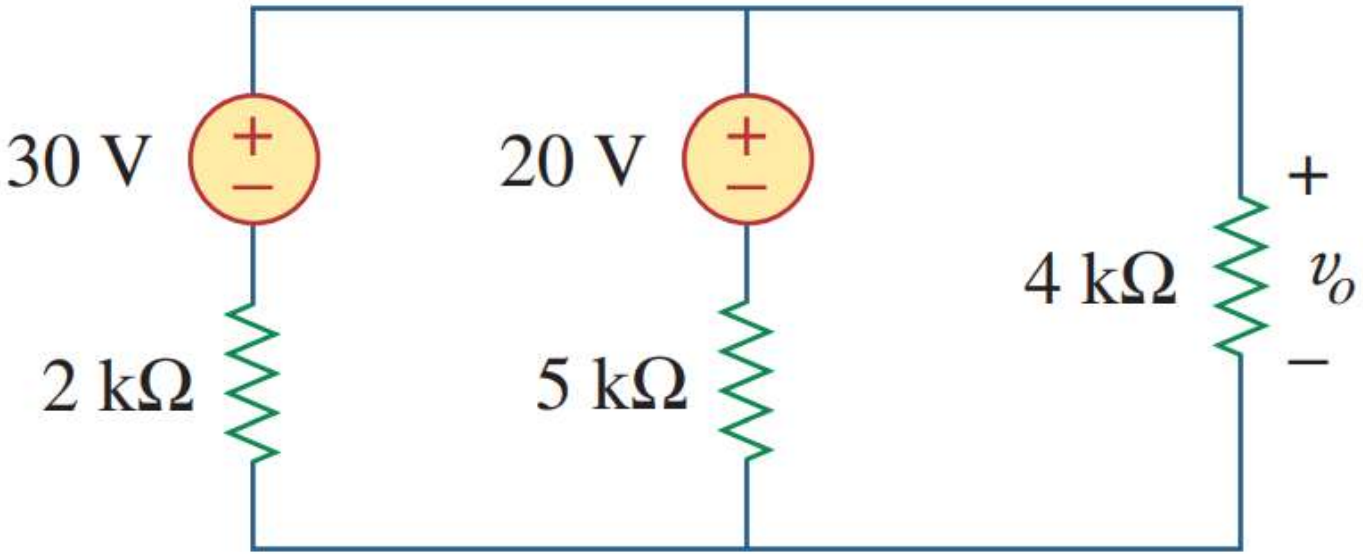
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Nodal Analysis

Steps to Determine Node Voltages:

1. Select a node as the reference node. Assign voltages v_1, $v_2, \ldots, v_{n-1}$ to the remaining $n - 1$ nodes. The voltages are referenced with respect to the reference node.
2. Apply KCL to each of the $n - 1$ nonreference nodes. Use Ohm's law to express the branch currents in terms of node voltages.
3. Solve the resulting simultaneous equations to obtain the unknown node voltages.

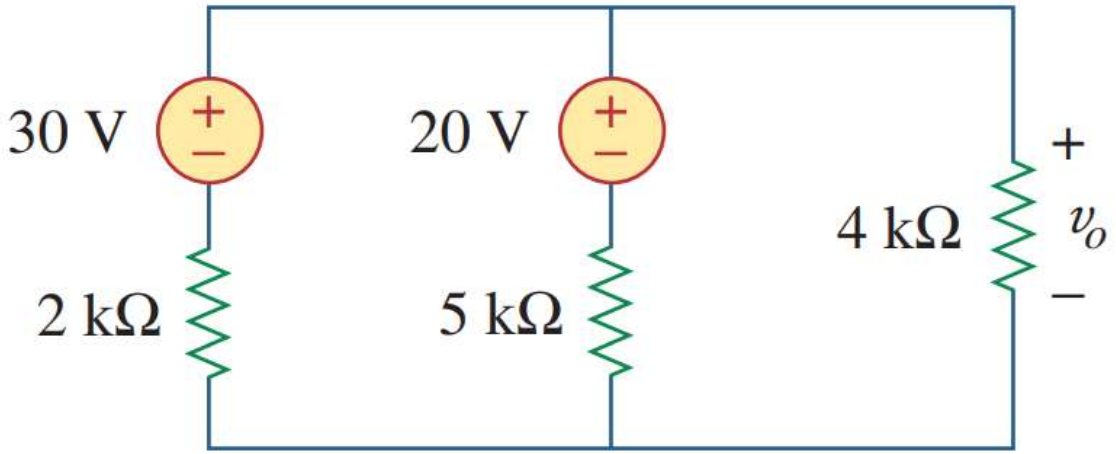
Nodal Analysis



Nodal Analysis

Basic equation

$$\frac{V_o - 30}{2000} + \frac{V_o - 20}{5000} + \frac{V_o}{4000} = 0$$



Formulated equation

$$\frac{V_o}{2000} + \frac{V_o}{5000} + \frac{V_o}{4000} = \frac{30}{2000} + \frac{20}{5000}$$

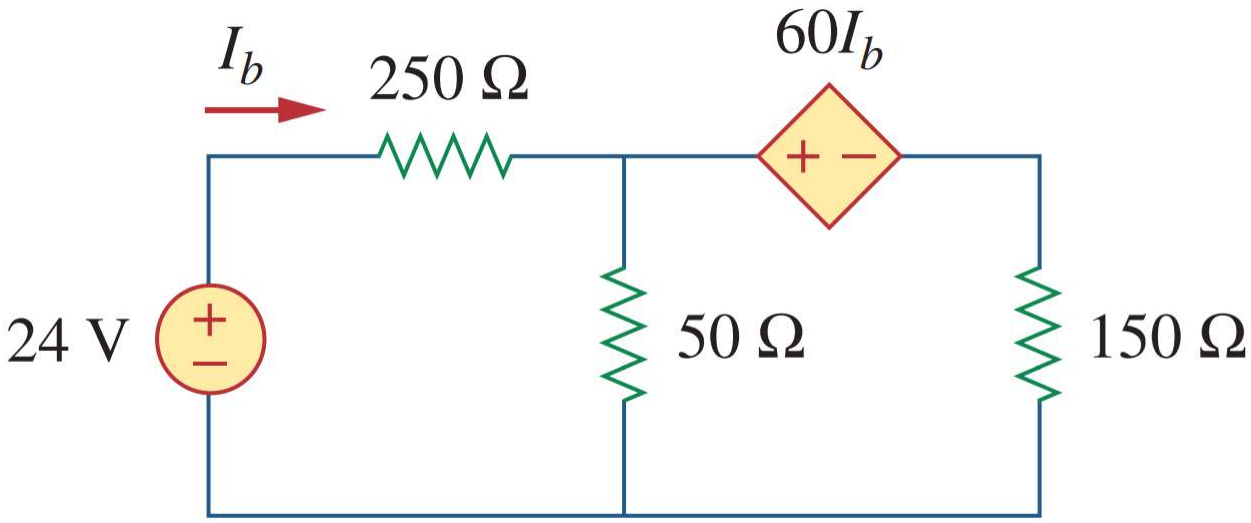
Final answer

$$V_o = 20\,V$$

Additional answer

$$XXXXXX$$

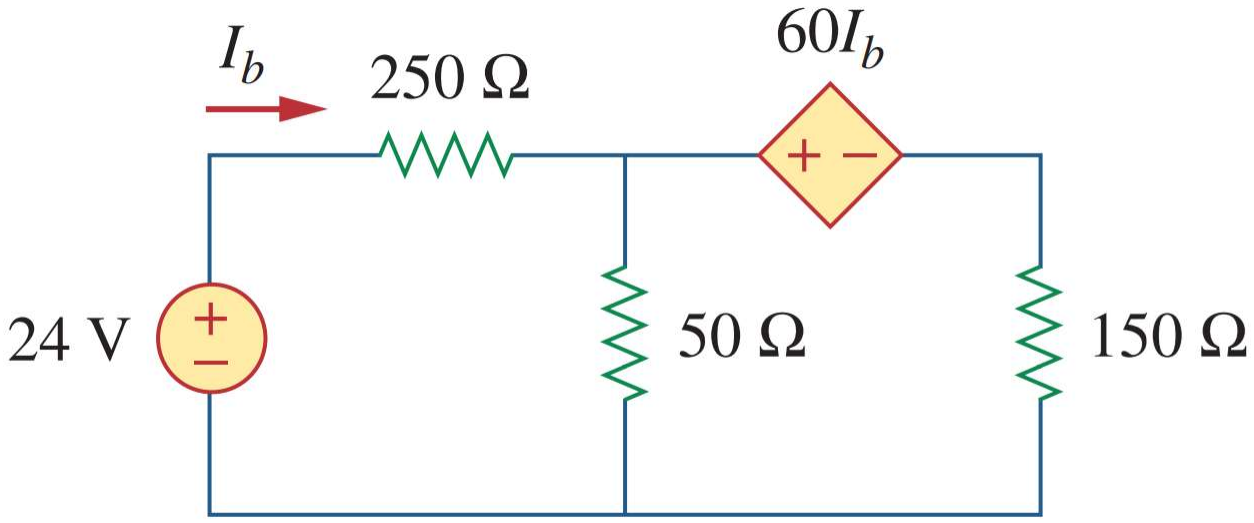
Nodal Analysis



Nodal Analysis

Basic equation

$$\frac{V_o - 24}{250} + \frac{V_o}{50} + \frac{V_o - 60(\frac{24 - V_o}{250})}{150} = 0$$



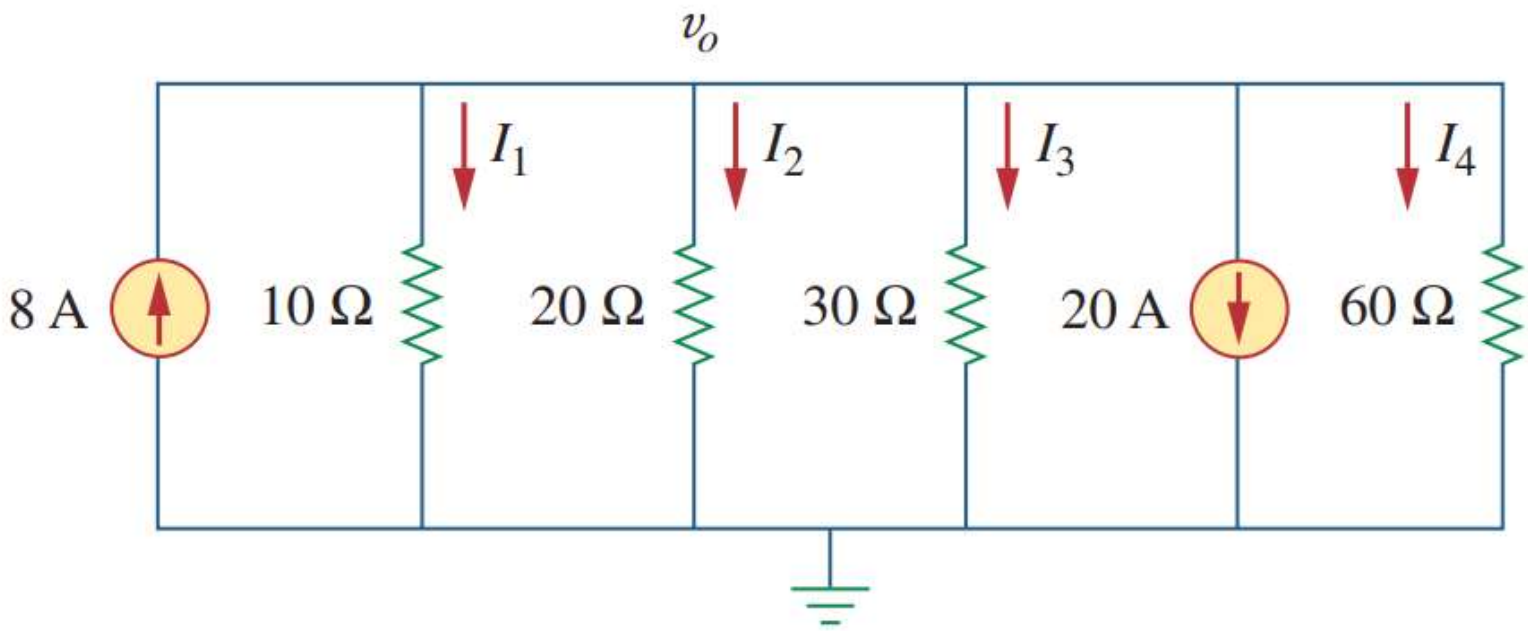
Formulated equation

$$\frac{V_o}{250} + \frac{V_o}{50} + \frac{V_o}{150} + \frac{(60 * V_o)/250}{150} = \frac{24}{250} + \frac{(60 * 24)/250}{150}$$

Final answer

$$V_o = 4.16\,V$$

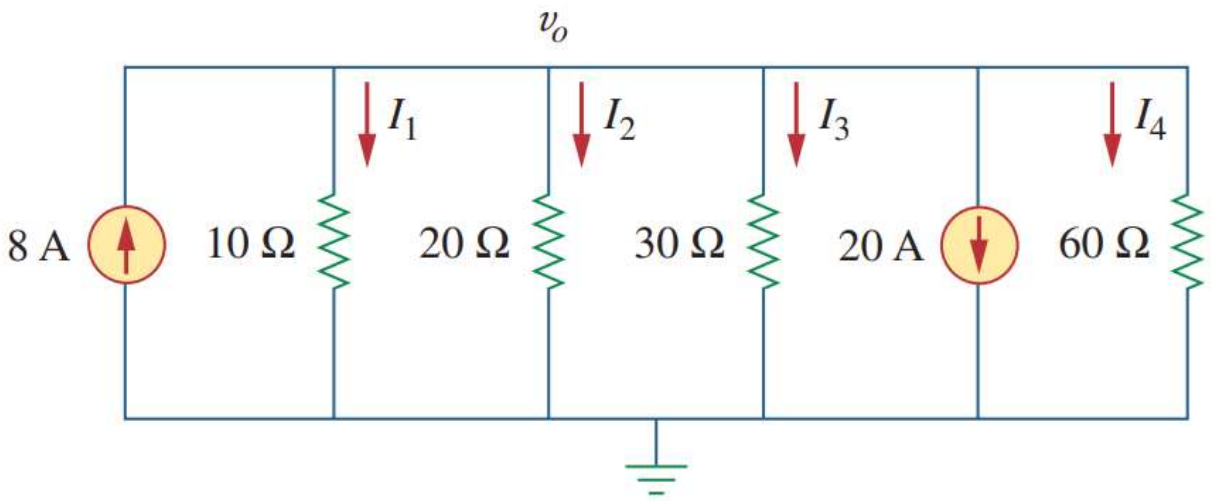
Nodal Analysis



Nodal Analysis

Basic equation

$$-8 + \frac{V_o}{10} + \frac{V_o}{20} + \frac{V_o}{30} + 20 - \frac{V_o}{60} = 0$$



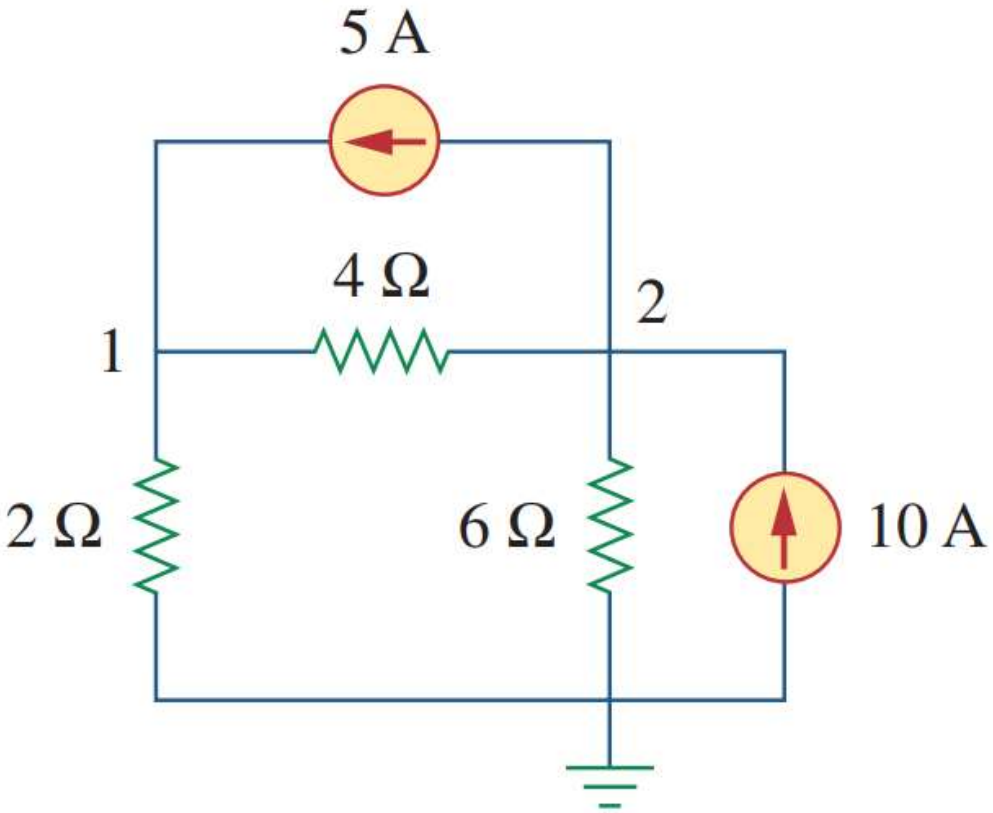
Formulated equation

$$\frac{V_o}{10} + \frac{V_o}{20} + \frac{V_o}{30} + \frac{V}{60} = -20 + 8$$

Final answer

$$V_o = -60\ V$$

Nodal Analysis



Nodal Analysis

Basic equation

$$\frac{V_1}{2} + \frac{V_1 - V_2}{4} - 5 = 0$$

$$\frac{V_2}{6} - 10 + \frac{V_2 - V_1}{4} + 5 = 0$$

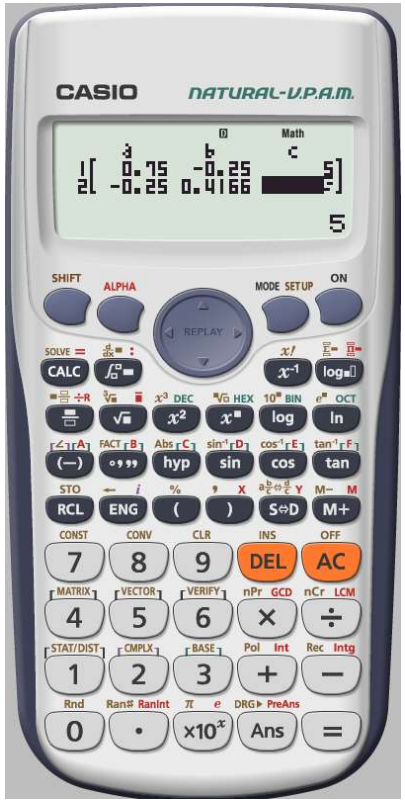
Formulated equation

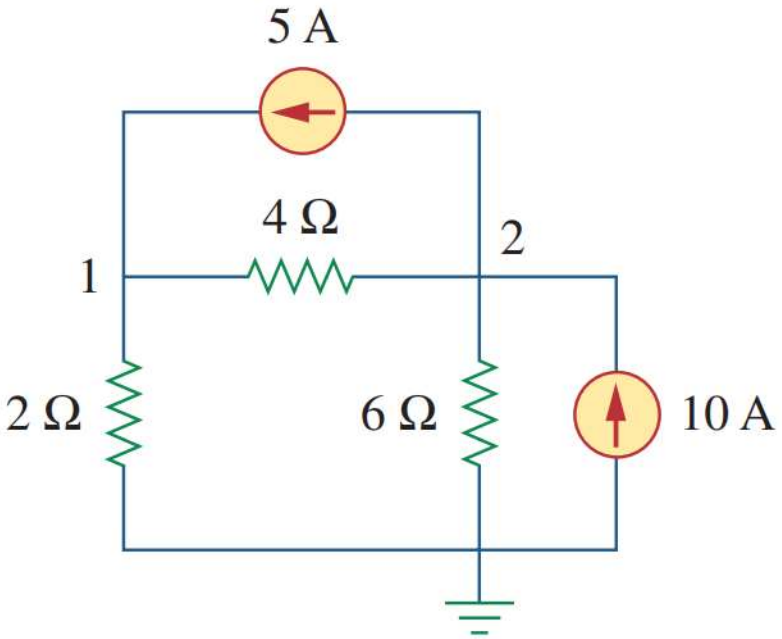
$$\left(\frac{1}{2} + \frac{1}{4}\right)V_1 - \frac{1}{4}V_2 = 5$$

$$\left(\frac{-1}{4}\right)V_1 + \left(\frac{1}{6} + \frac{1}{4}\right)V_2 = 5$$

Final answer

$V_1 = 13.333\ V$ $V_2 = 20\ V$







Electrical Circuit (1)

Introduction (week3 class2)

Dr. Akram Al-Mahrouk

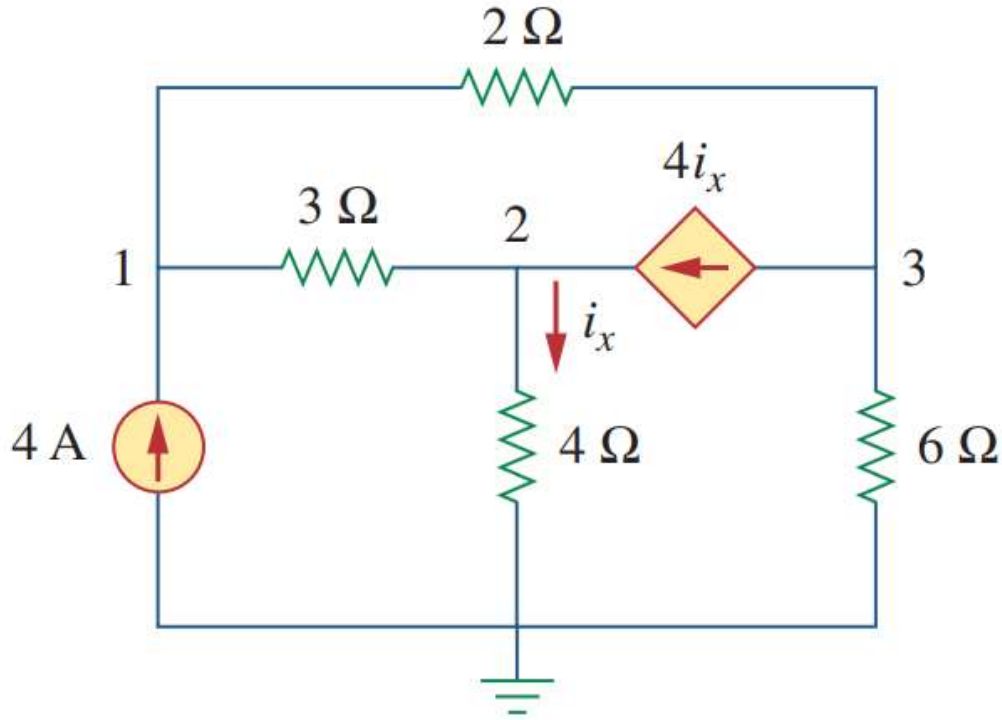
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Advanced

Nodal Analysis



Nodal Analysis

Basic equation

$$-4+\frac{V_1-V_2}{3}+\frac{V_1-V_3}{2}=0$$

$$\frac{V_2-V_1}{3}+\frac{V_2}{4}-4\frac{V_2}{4}=0$$

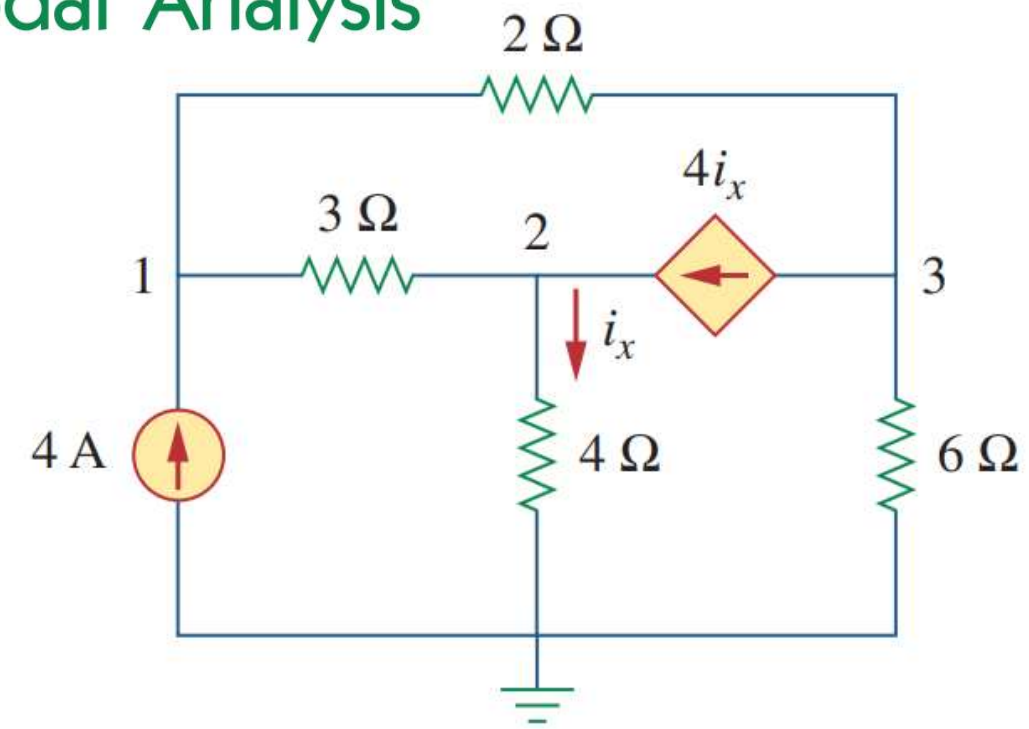
$$\frac{V_3-V_1}{2}+\frac{V_3}{6}+4\frac{V_2}{4}=0$$

Formulated equation

$$\left(\frac{1}{3}+\frac{1}{2}\right)V_1-\left(\frac{1}{3}\right)V_2-\left(\frac{1}{2}\right)V_3=4$$

$$\left(\frac{-1}{3}\right)V_1+\left(\frac{1}{3}+\frac{1}{4}-\frac{4}{4}\right)V_2=0$$

$$\left(\frac{-1}{2}\right)V_1+\left(\frac{4}{4}\right)V_2+\left(\frac{1}{2}+\frac{1}{6}\right)V_3=0$$



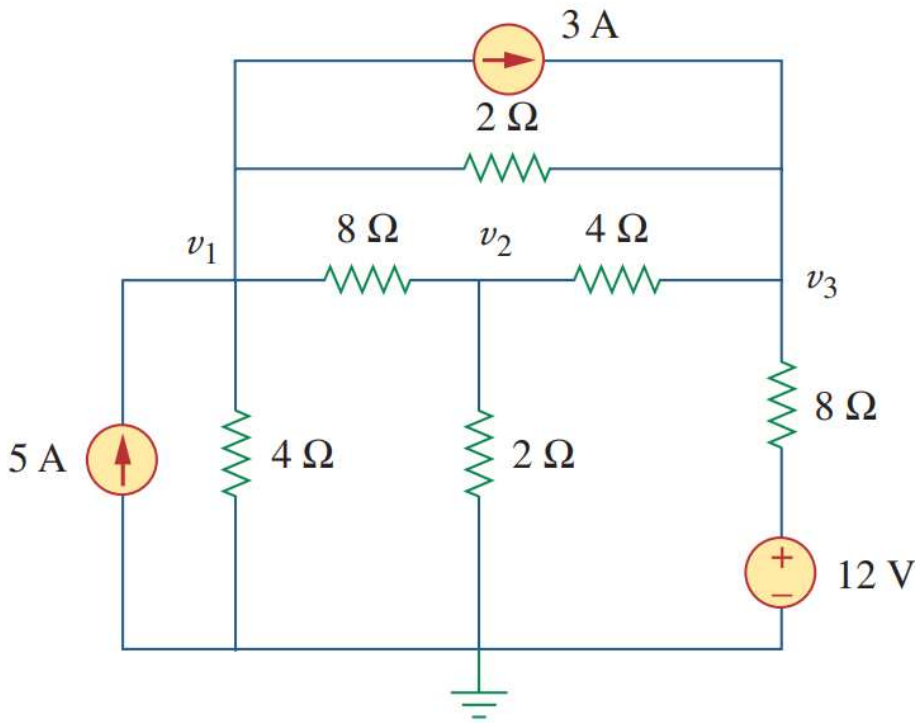
Final answer

$$V_1 = 32\ V$$

$$V_2 = -25.6\ V$$

$$V_3 = 62.4\ V$$

Nodal Analysis



Nodal Analysis

Basic equation

$$-5+\frac{V_1}{4}+\frac{V_1-V_2}{8}+\frac{V_1-V_3}{2}+3=0$$

$$\frac{V_2-V_1}{8}+\frac{V_2-V_3}{4}+\frac{V_2}{2}=0$$

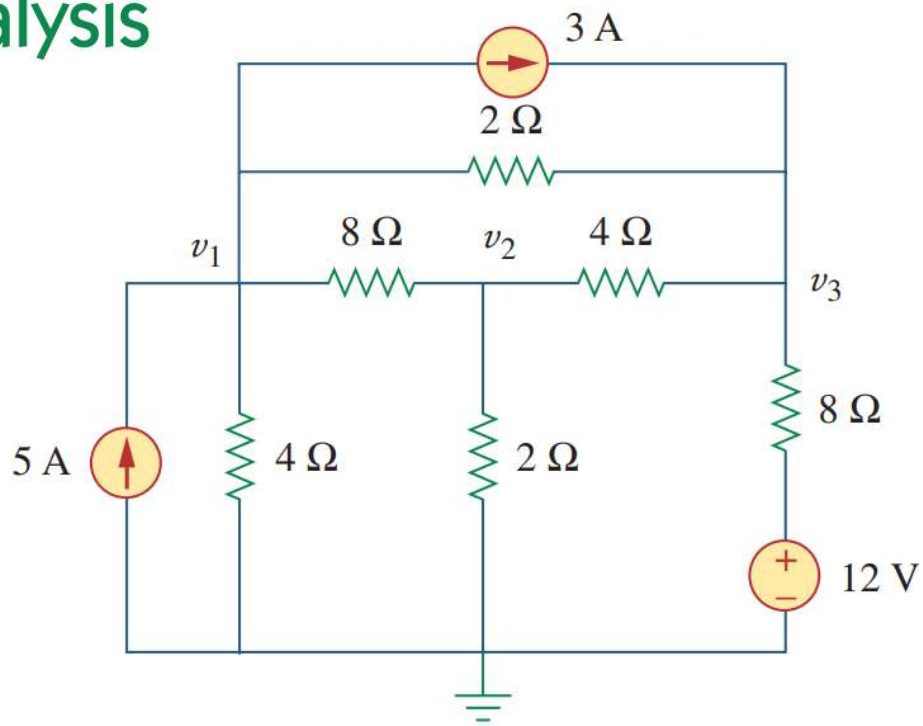
$$-3+\frac{V_3-V_1}{2}+\frac{V_3-V_2}{4}+\frac{V_3-12}{8}=0$$

Formulated equation

$$\left(\frac{1}{4}+\frac{1}{8}+\frac{1}{2}\right)V_1-\left(\frac{1}{8}\right)V_2-\left(\frac{1}{2}\right)V_3=5-3$$

$$\left(\frac{-1}{8}\right)V_1+\left(\frac{1}{8}+\frac{1}{4}+\frac{1}{2}\right)V_2+\left(\frac{-1}{4}\right)V_1=0$$

$$\left(\frac{-1}{2}\right)V_1+\left(-\frac{1}{4}\right)V_2+\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}\right)V_3=3+\frac{12}{8}$$



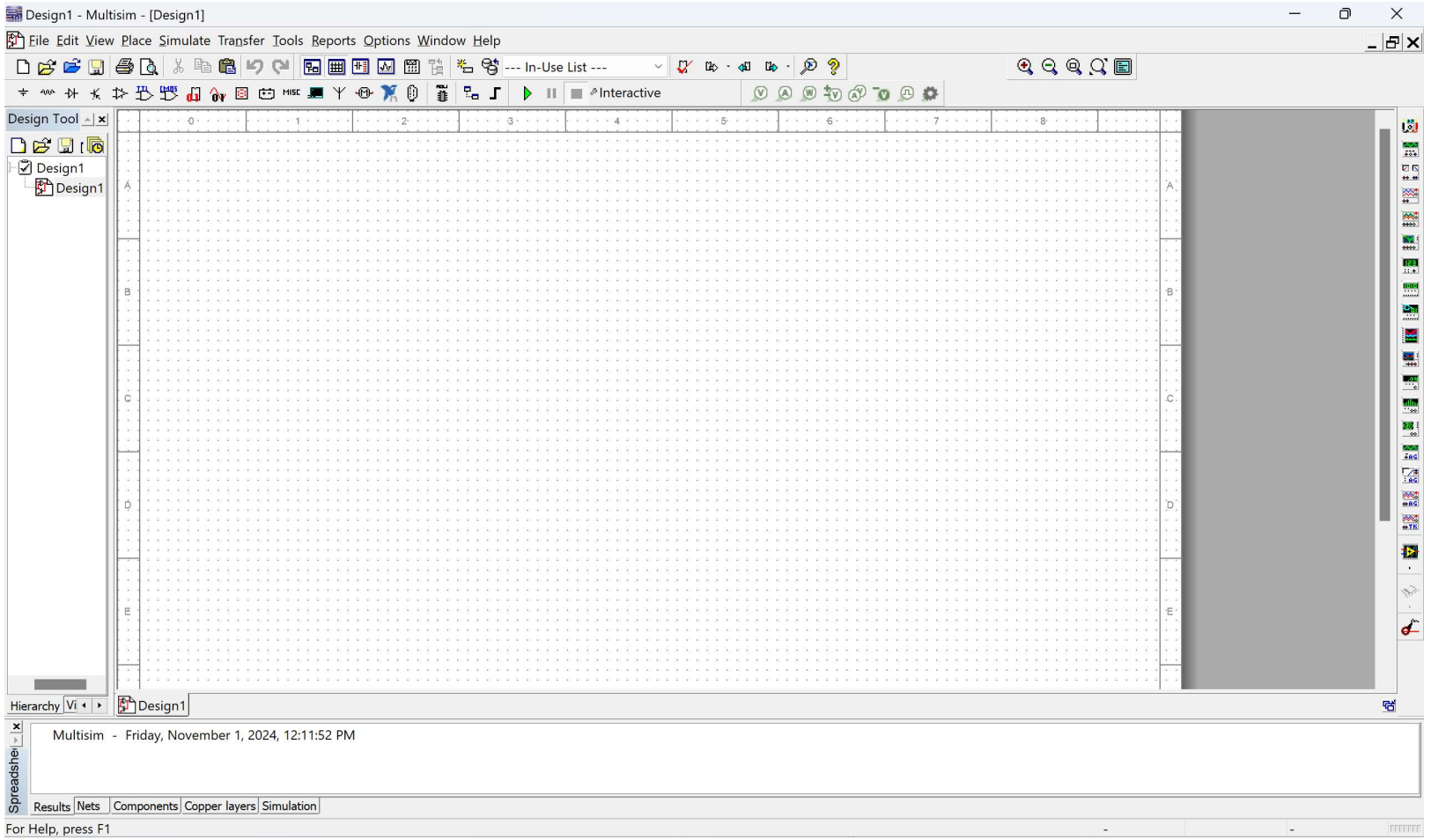
Final answer

$$V_1 = 10\ V$$

$$V_2 = 4.9333\ V$$

$$V_3 = 12.2666\ V$$


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File Edit View Place Simulate Transfer Tools Reports Options Window Help
--- In-Use List ---
Interactive
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Hierarchy
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Spreadsheet
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Electrical Circuit (1)

Introduction (week4 class1)

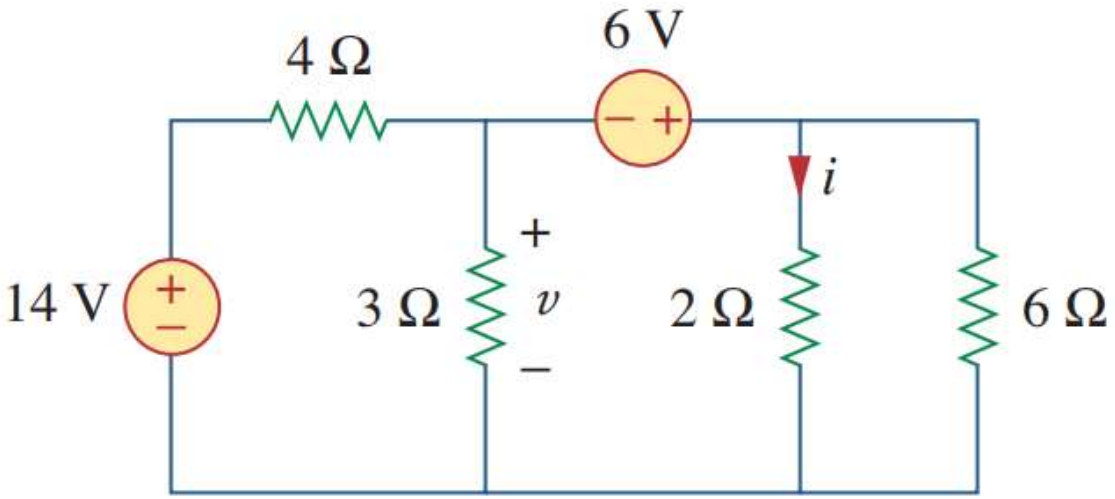
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Nodal Analysis with Voltage Sources





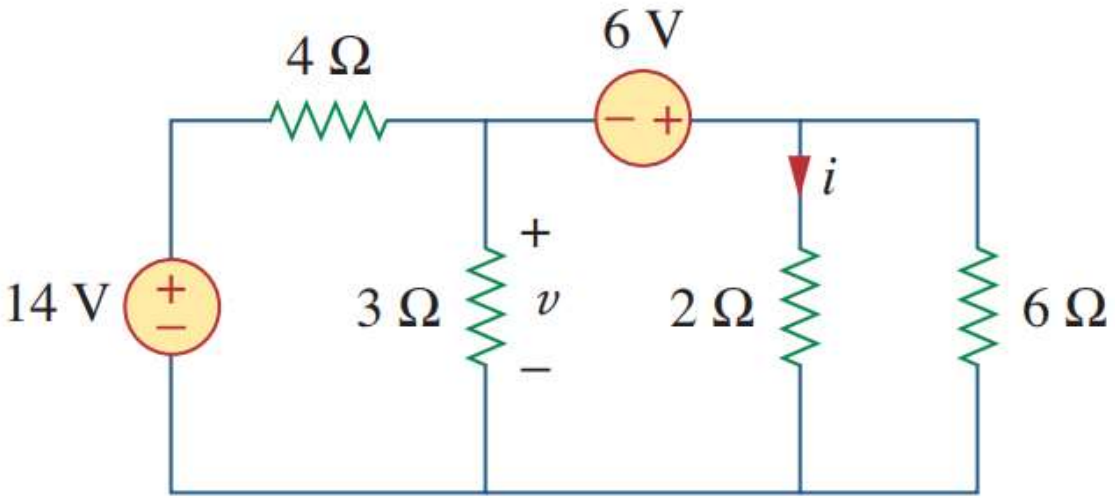
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Nodal Analysis with Voltage Sources

Basic equation

$$V_1 + 6 = V_2$$

$$\frac{V_1 - 14}{4} + \frac{V_1}{3} + \frac{V_2}{2} + \frac{V_2}{6} = 0$$



Formulated equation

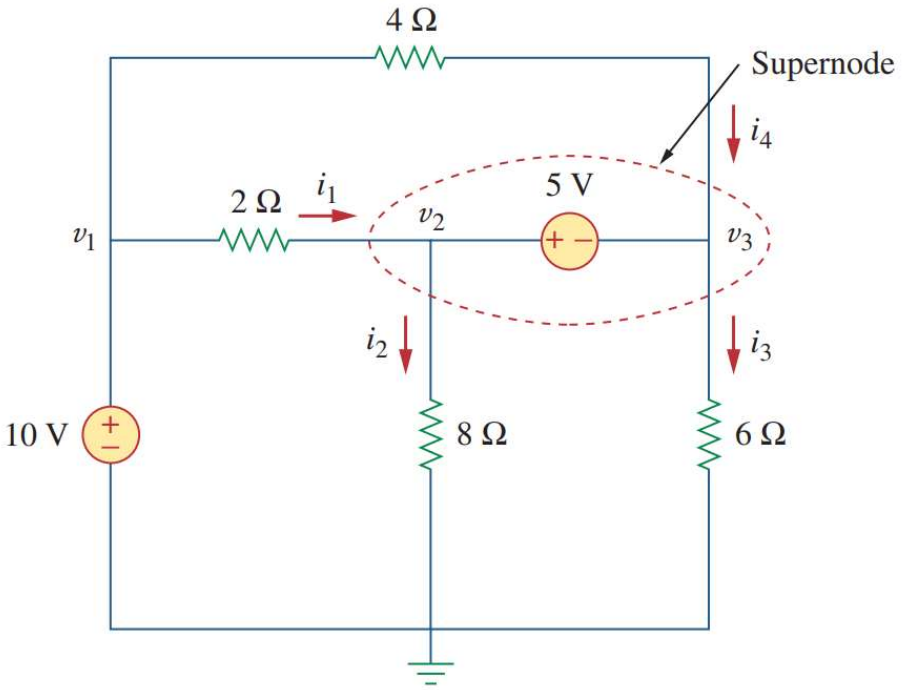
$$V_1 - V_2 = -6$$

$$\left(\frac{1}{4} + \frac{1}{3}\right)V_1 + \left(\frac{1}{2} + \frac{1}{6}\right)V_2 = \frac{14}{4}$$

Final answer

$$V_1 = -0.4\,mV \qquad V_2 = 5.6\,V$$

Nodal Analysis with Voltage Sources



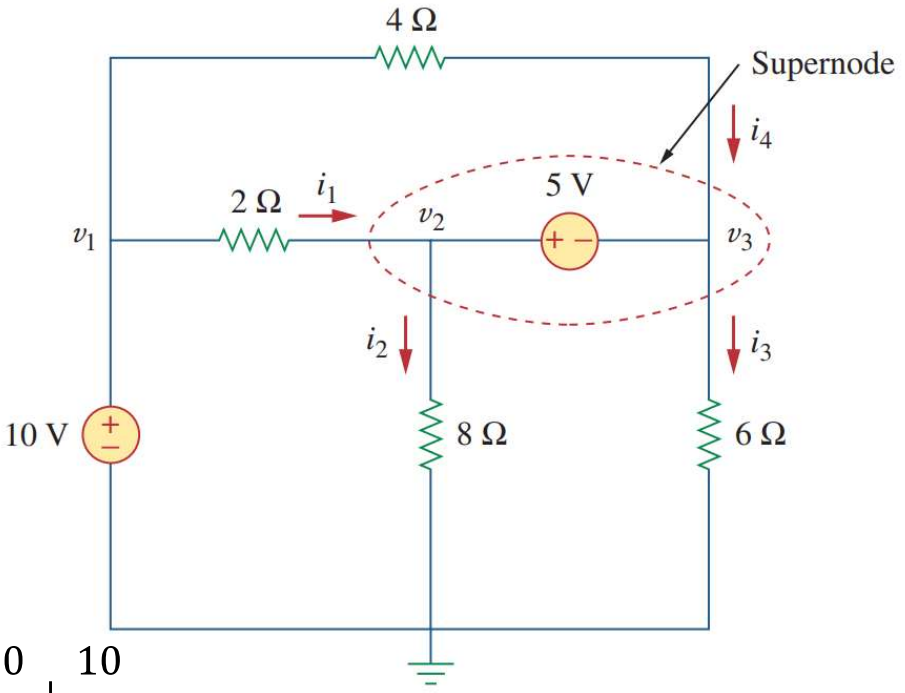
Nodal Analysis with Voltage Sources

Basic equation

$V_1 = 10$

$V_3 + 5 = V_2$

$$\frac{V_2 - V_1}{2} + \frac{V_2}{8} + \frac{V_3 - V_1}{4} + \frac{V_3}{6} = 0$$



Formulated equation

$V_2 - V_3 = 5$

$$\frac{V_2 - 10}{2} + \frac{V_2}{8} + \frac{V_3 - 10}{4} + \frac{V_3}{6} = 0 \qquad \left(\frac{1}{2} + \frac{1}{8}\right)V_2 + \left(\frac{1}{4} + \frac{1}{6}\right)V_3 = \frac{10}{2} + \frac{10}{4}$$

Final answer

$V_1 = 10\ V$ $V_2 = 9.2\ V$ $V_3 = 4.2\ V$



Electrical Circuit (1)

Introduction (week3 class2)

Dr. Akram Al-Mahrouk

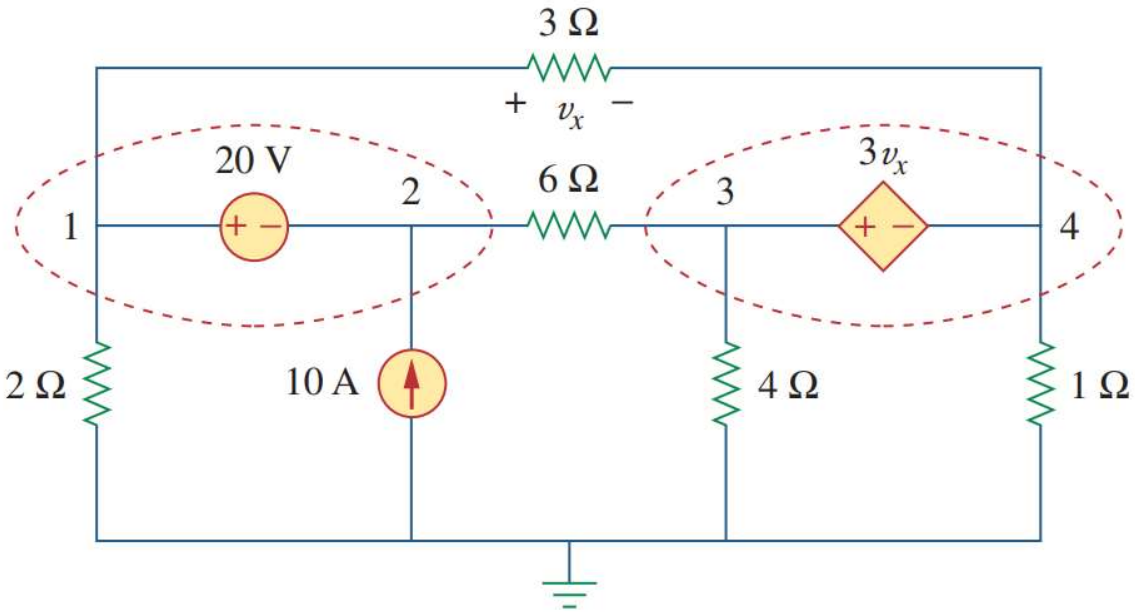
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Nodal Analysis with Voltage Sources



Nodal Analysis with Voltage Sources

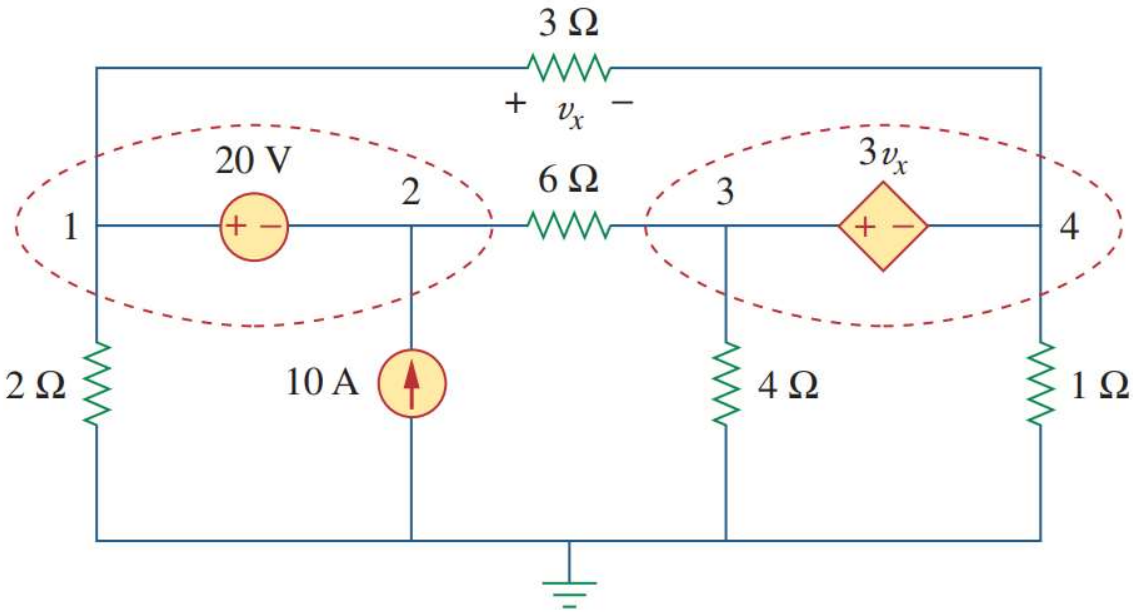
Basic equation

$$V_1 = V_2 + 20$$

$$V_3 = V_4 + 3(V_1 - V_4)$$

$$\frac{V_1}{2} + \frac{V_1 - V_4}{3} + (-10) + \frac{V_2 - V_3}{6} = 0$$

$$\frac{V_3}{4} + \frac{V_3 - V_2}{6} + \frac{V_4}{1} + \frac{V_4 - V_1}{3} = 0$$



Formulated equation

$$V_1 - V_2 = 20$$

$$3V_1 - V_3 - 2V_4 = 0$$

$$\left(\frac{1}{2} + \frac{1}{3}\right)V_1 + \left(\frac{1}{6}\right)V_2 + \left(\frac{-1}{6}\right)V_3 + \left(\frac{-1}{3}\right)V_4 = 10$$

$$\frac{-1}{3}V_1 + \frac{-1}{6}V_2 + (\frac{1}{4} + \frac{1}{6})V_3 + (1 + \frac{1}{3})V_4 = 0$$

Final answer

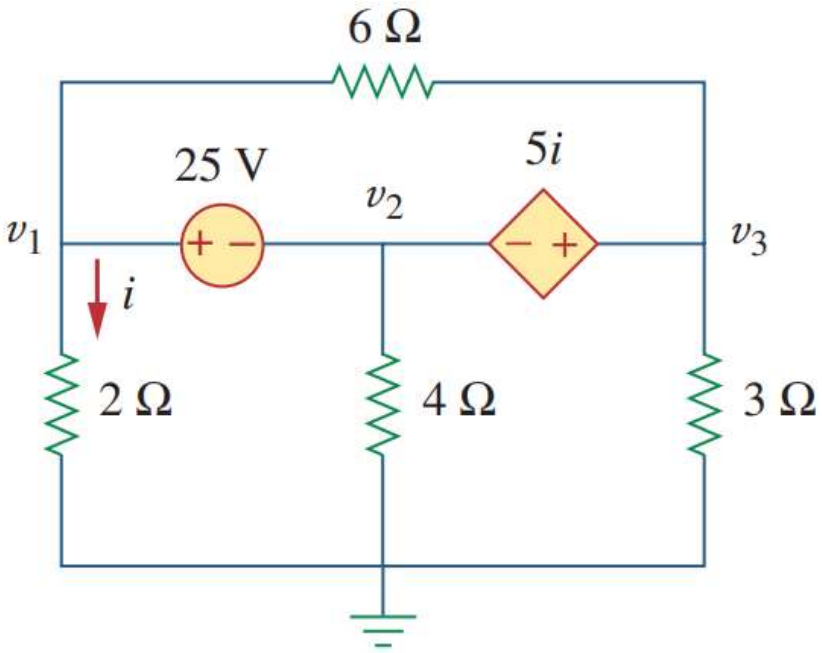
$$V_1 = -26.6667\ V$$

$$V_2 = 6.66667\ V$$

$$V_3 = 173.333\ V$$

$$V_4 = -46.6667\ V$$

Nodal Analysis with Voltage Sources



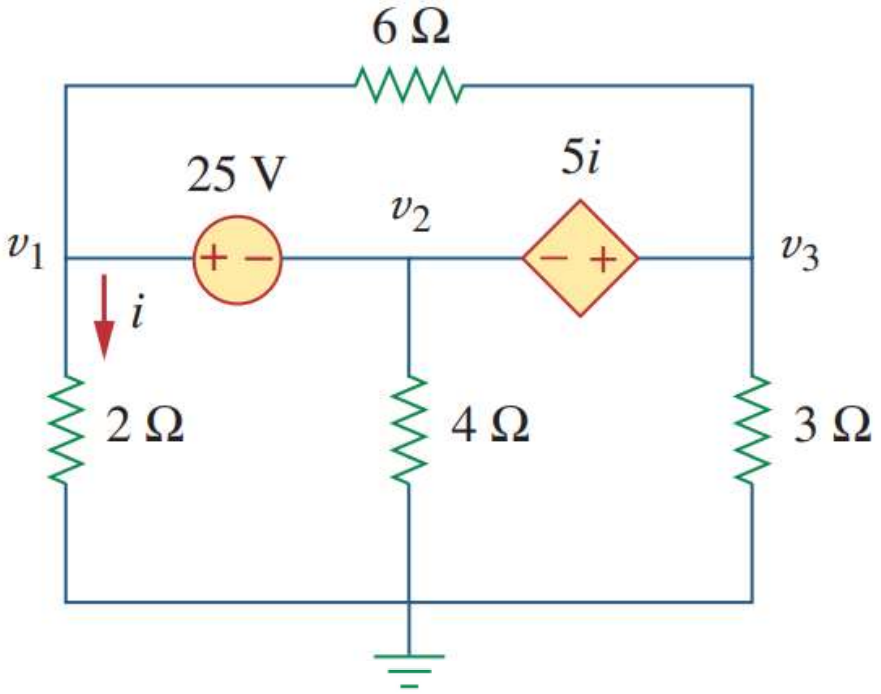
Nodal Analysis with Voltage Sources

Basic equation

$V_1 = V_2 + 25$

$V_3 = V_2 + 5(\frac{V_1}{2})$

$\frac{V_1}{2} + \frac{V_2}{4} + \frac{V_3}{3} = 0$



Formulated equation

$V_1 - V_2 = 25$

$(\frac{5}{2})V_1 + V_2 - V_3 = 0$

$\left(\frac{1}{2}\right)V_1 + \left(\frac{1}{4}\right)V_2 + \left(\frac{1}{3}\right)V_3 = 0$

Final answer

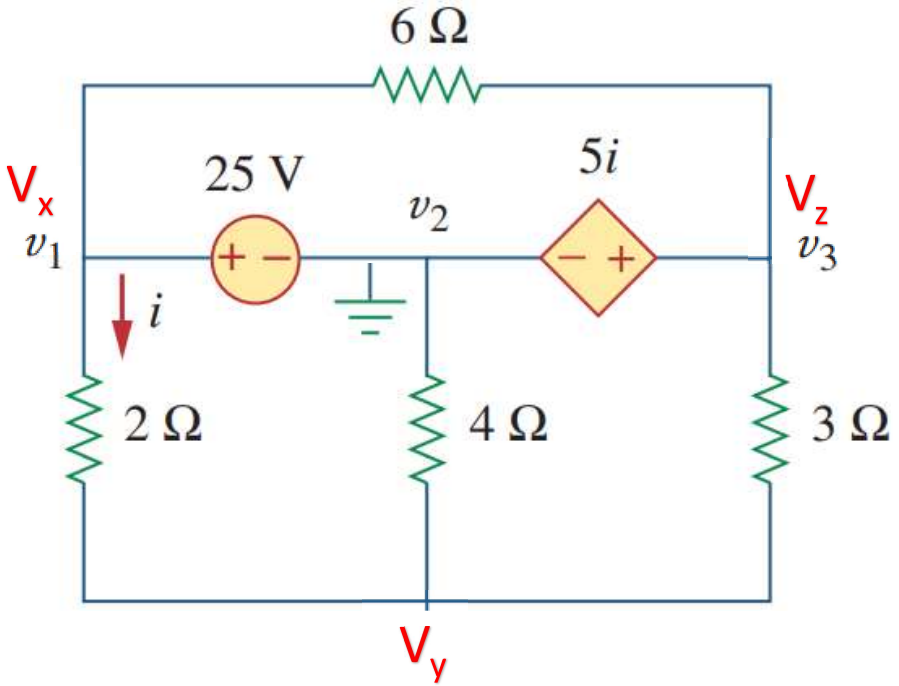
$V_1 = 7.608\ V$

$V_2 = -17.39\ V$

$V_3 = 1.630\ V$

Nodal Analysis with Voltage Sources

Without super node



Nodal Analysis with Voltage Sources

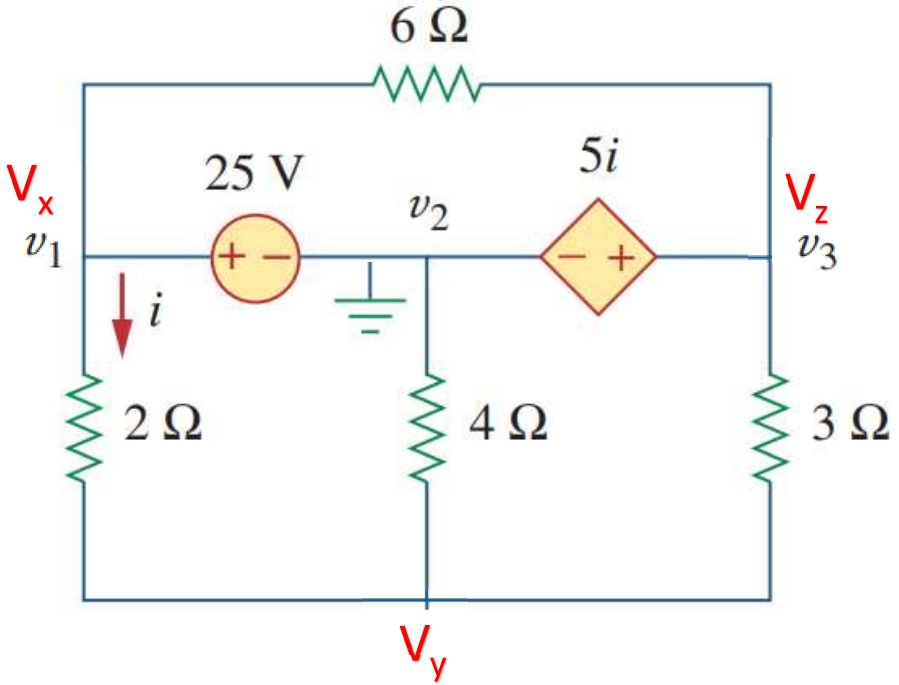
Without super node

Basic equation

$$V_x = 25$$

$$V_z = 5\left(\frac{V_x - V_y}{2}\right) \Rightarrow V_z = 5\left(\frac{25 - V_y}{2}\right)$$

$$\frac{V_y - V_x}{2} + \frac{V_y}{4} + \frac{V_y - V_z}{3} = 0 \Rightarrow \frac{V_y - 25}{2} + \frac{V_y}{4} + \frac{V_y - V_z}{3} = 0$$



Formulated equation

$$\left(\frac{5}{2}\right)V_y + V_z = \left(\frac{5 * 25}{2}\right)$$

$$\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{3}\right)V_y + \frac{-1}{3}V_z = \frac{25}{2}$$

Final answer

$$V_x = 25\ V \qquad V_1 = V_x - V_y = 25 - 17.39 = 7.608\ V$$

$$V_y = 17.39\ V \qquad V_2 = -V_y = -17.39\ V$$

$$V_z = 19.021\ V \qquad V_3 = V_z - V_y = 19.021 - 17.39 = 1.630\ V$$



Electrical Circuit (1)

Introduction (week5 class1)

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Mesh Analysis

Steps to Determine Mesh Currents:

1. Assign mesh currents $i_1, i_2, \ldots, i_n$ to the n meshes.
2. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting n simultaneous equations to get the mesh currents.

Mesh Analysis

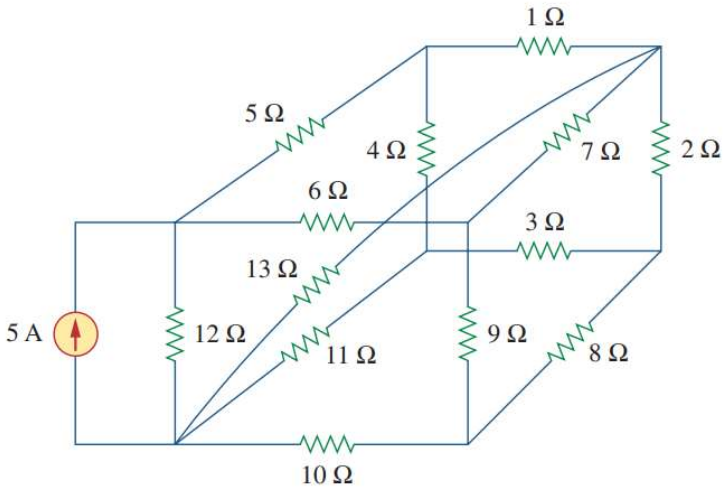
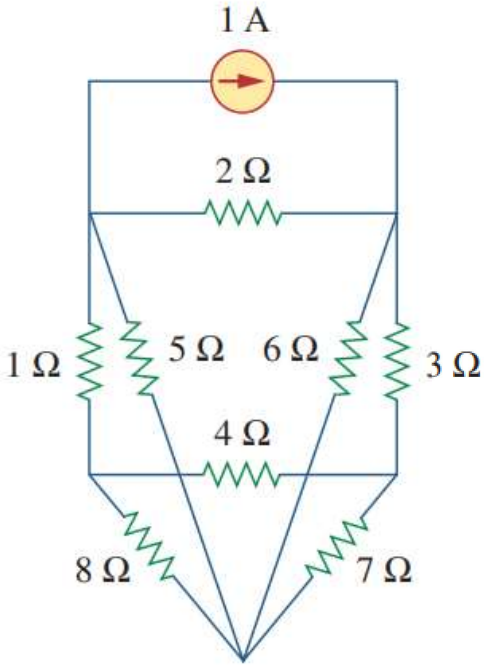
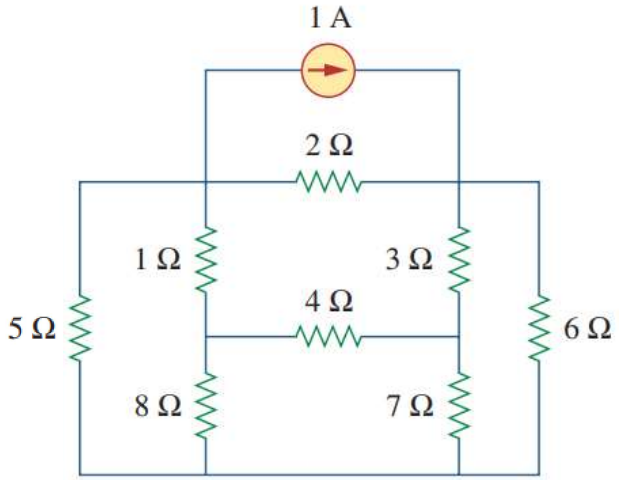


Figure 3.16
A nonplanar circuit.



=



Mesh Analysis

Basic equation

$$-15 + (5 * i_1) + 10 * (i_1 - i_2) + 10 = 0$$

$$-10 + 10 * (i_2 - i_1) + (6 * i_2) + (4 * i_2) = 0$$

Formulated equation

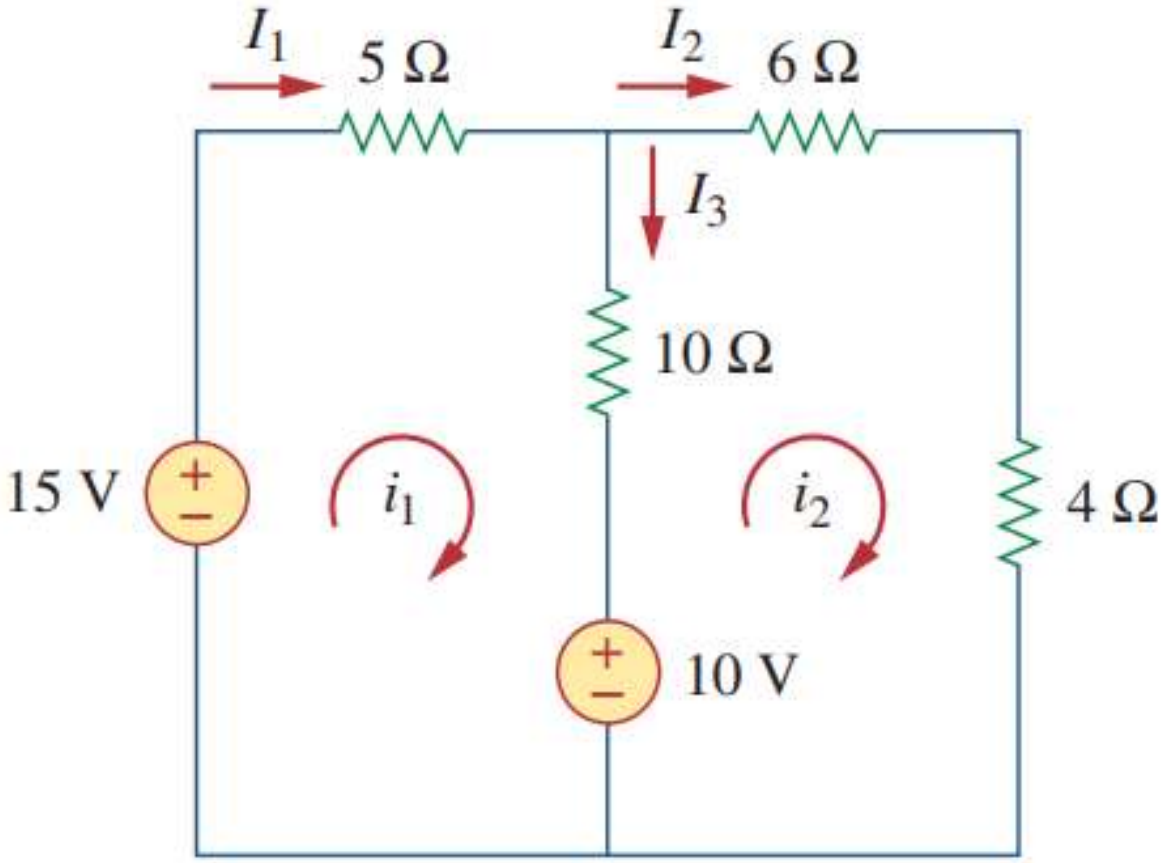
$$15i_1 - 10i_2 = 5$$

$$-10i_1 + 20i_2 = 10$$

Final answer

$$i_1 = 1\ \text{A}$$

$$i_2 = 1\ \text{A}$$



Mesh Analysis

Basic equation

$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0$$

$$10(i_2 - i_1) + 24i_2 + 4(i_2 - i_3) = 0$$

$$12(i_3 - i_1) + 4(i_3 - i_2) + 4(i_1 - i_2) = 0$$

Formulated equation

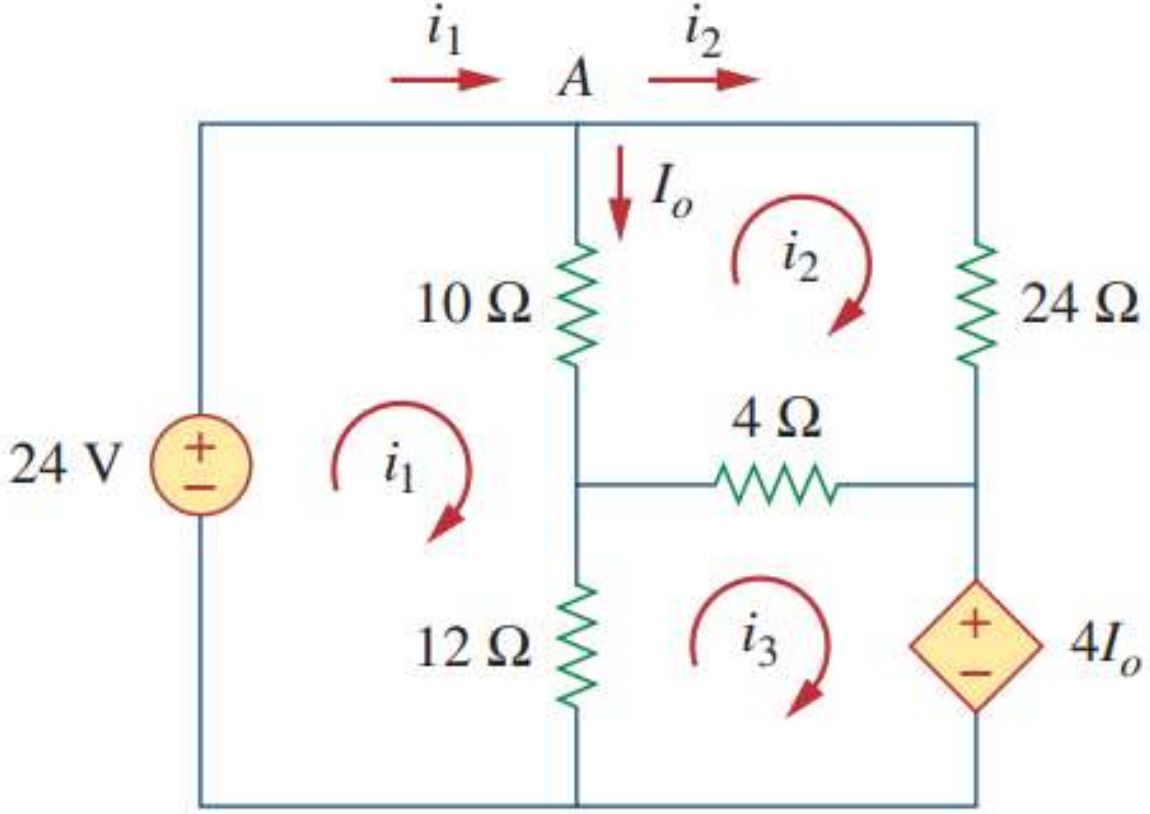
$$22i_1 - 10i_2 - 12i_3 = 24$$

$$-10i_1 + 38i_2 - 4i_3 = 0$$

$$-8i_1 - 8i_2 + 16i_3 = 0$$

Final answer

$i_1 = 2.25A$ $\quad$ $i_2 = 0.75\ A$ $\quad$ $i_3 = 1.5\ A$





Electrical Circuit (1)

Introduction (week5 class2)

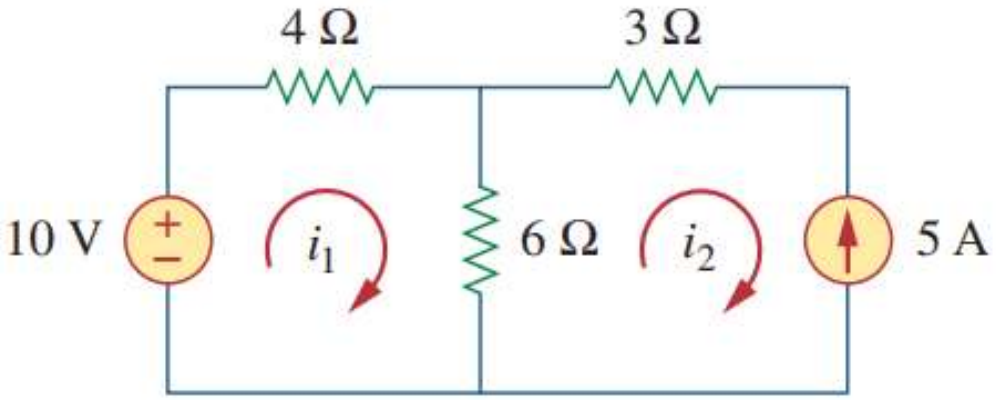
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Mesh Analysis with Current Sources

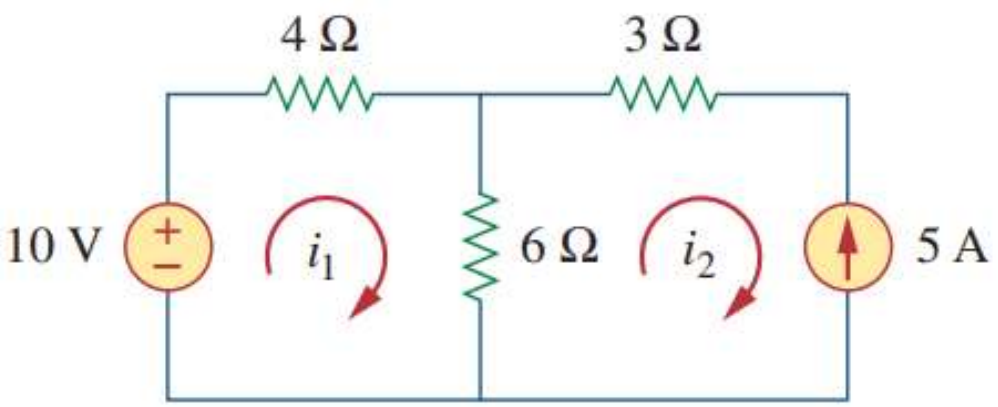


Mesh Analysis with Current Sources

Basic equation

$$-10 + (4 * i_1) + 6 * (i_1 - i_2) = 0$$

$$i_2 = -5$$

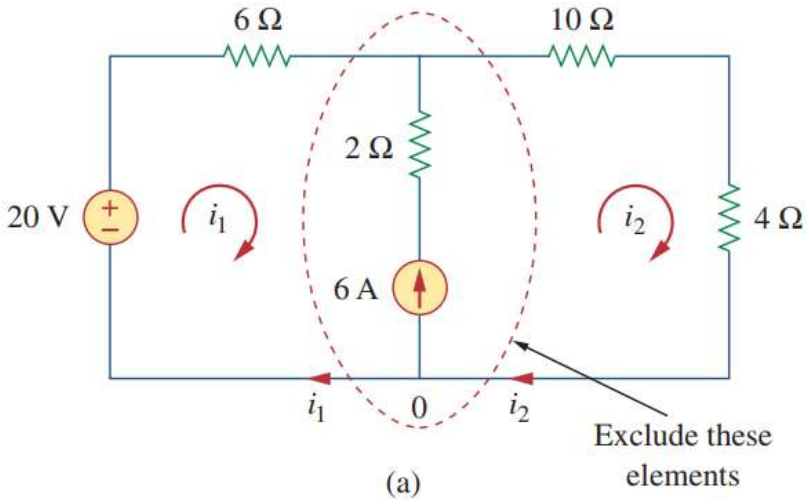


Final answer

$$i_1 = -2 \text{ A}$$

$$i_2 = -5 \text{ A}$$

Mesh Analysis with Current Sources



Mesh Analysis with Current Sources

Basic equation

$$i_1 + 6 = i_2$$

$$-20 + (6 * i_1) + (10 * i_2) + (4 * i_2) = 0$$

Formulated equation

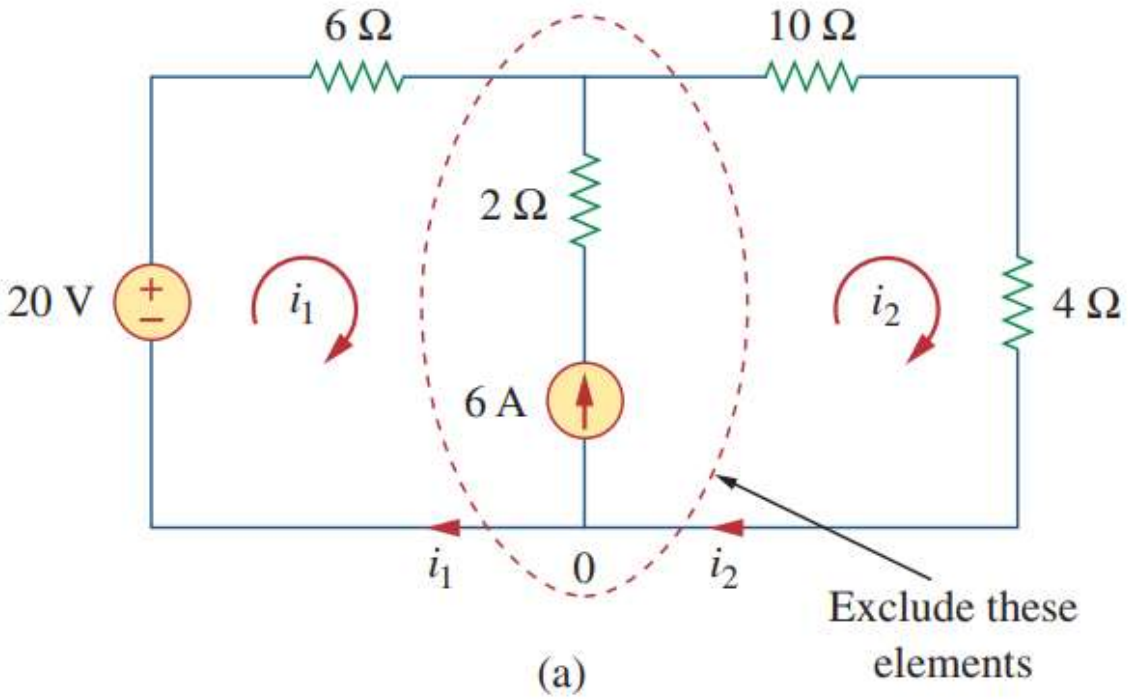
$$i_1 - i_2 = -6$$

$$6i_1 + 14i_2 = 20$$

Final answer

$$i_1 = -3.2 \text{ A}$$

$$i_2 = 2.8 \text{ A}$$





Electrical Circuit (1)

Circuit Technique (week6 class1)

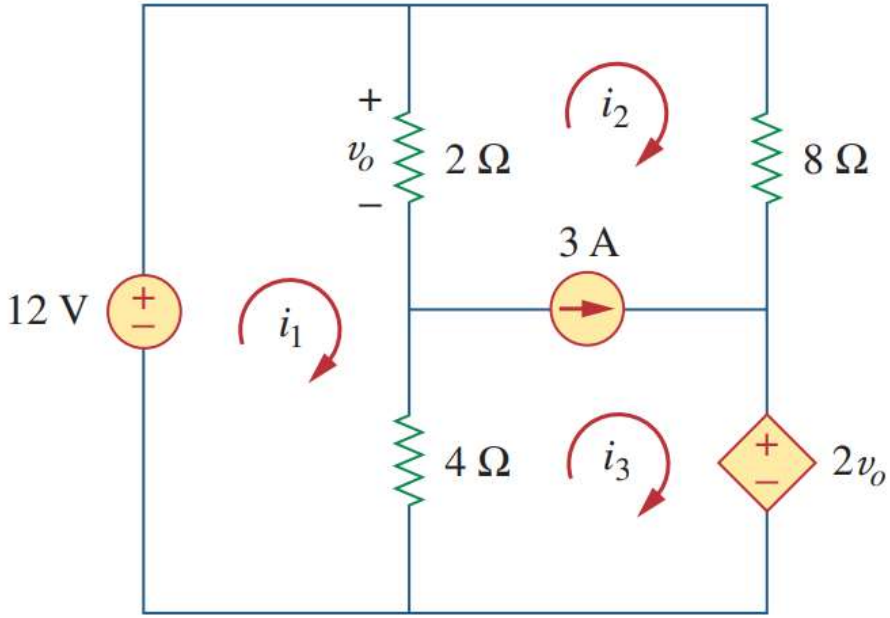
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Mesh Analysis with Current Sources



Mesh Analysis with Current Sources

Basic equation

$$-12 + 2(i_1 - i_2) + 4(i_1 - i_3) = 0$$

$$i_3 - i_2 = 3$$

$$4(i_3 - i_1) + 2(i_2 - i_1) + 8i_2 + 4(i_1 - i_2)) = 0$$

Formulated equation

$$6i_1 - 2i_2 - 4i_3 = 12$$

$$-i_2 + i_3 = 3$$

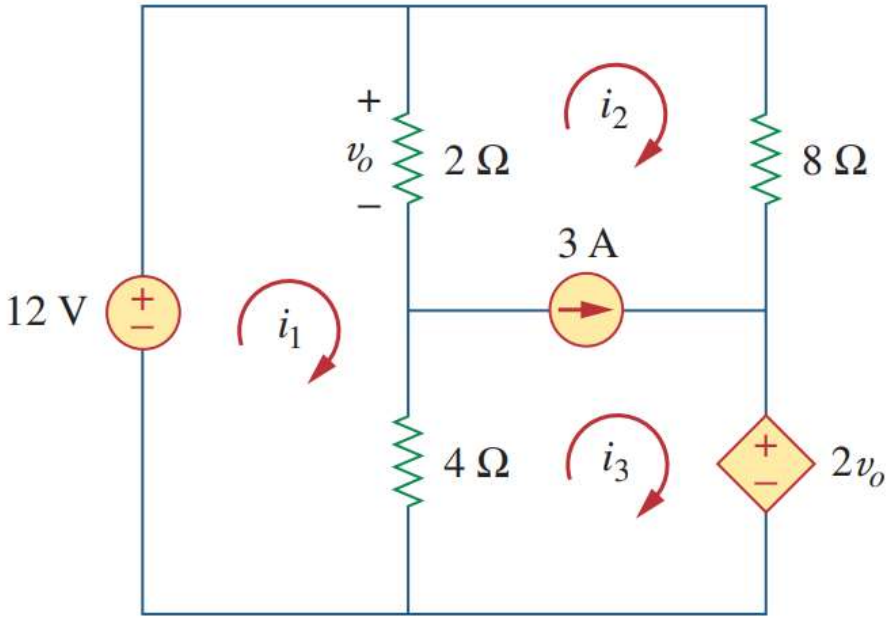
$$-2i_1 + 6i_2 + 4i_3 = 0$$

Final answer

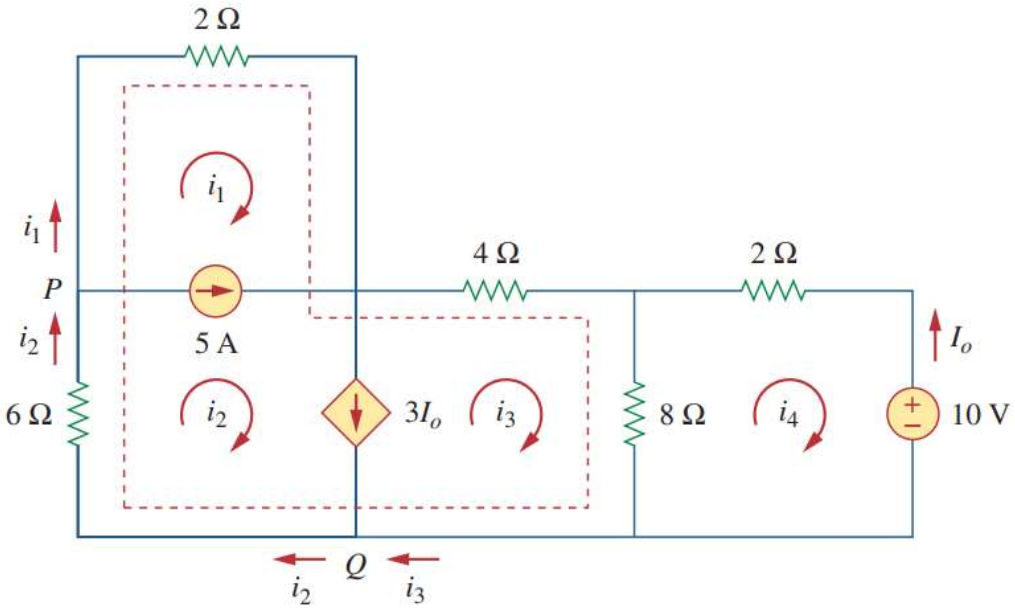
$$i_1 = 3.5 \text{ A}$$

$$i_2 = -0.5 \text{ A}$$

$$i_3 = 2.5 \text{ A}$$



Mesh Analysis with Current Sources



Mesh Analysis with Current Sources

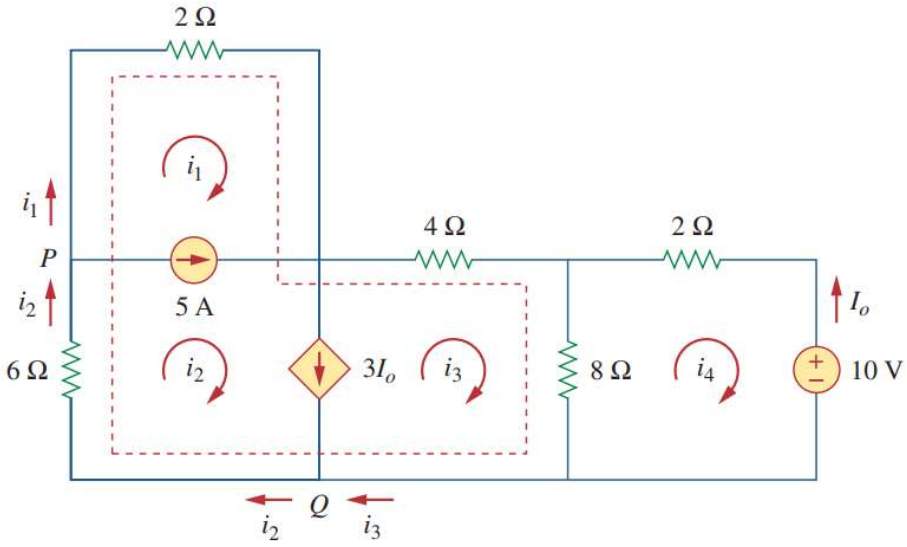
Basic equation

$i_2 - i_1 = 5$

$i_2 - i_3 = 3i_0 \qquad i_0 = -i_4$

$6i_2 + 2i_1 + 4i_3 + 8(i_3 - i_4) = 0$

$8(i_4 - i_3) + 2i_4 + 10 = 0$



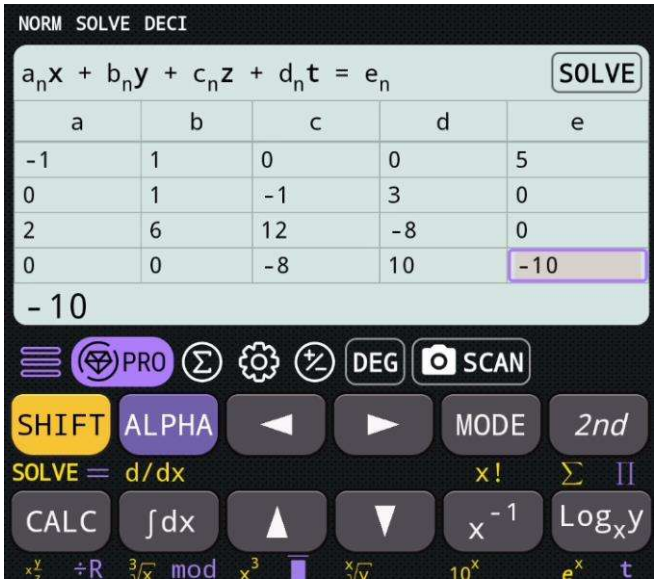
Formulated equation

$-i_1 + i_2 = 5$

$i_2 - i_3 + 3i_4 = 0$

$2i_1 + 6i_2 + 12i_3 - 8i_4 = 0$

$-8i_3 + 10i_4 = -10$



Final answer

$i_1 = -7.5$ A

$i_2 = -2.5$ A

$i_3 = 3.9286$ A

$i_4 = 2.1429$ A

